



Data Converters

Chapter 04

Digital-to-Analog Converters

Summer Term 2026

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VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

during his sabbatical at the University of Freiburg in 2009



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Overview

- **Types**
 - **Thermometer**
 - Binary Weighted
 - Segmented
- Static Performance
- Architectures

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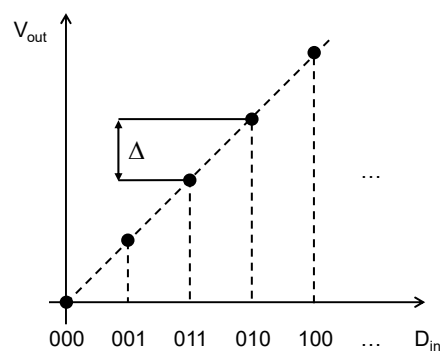
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Ideal Digital-to-Analog Converter - DAC

- essentially a digitally controlled voltage, current, or charge source
 - no quantization error!



example: unipolar DAC

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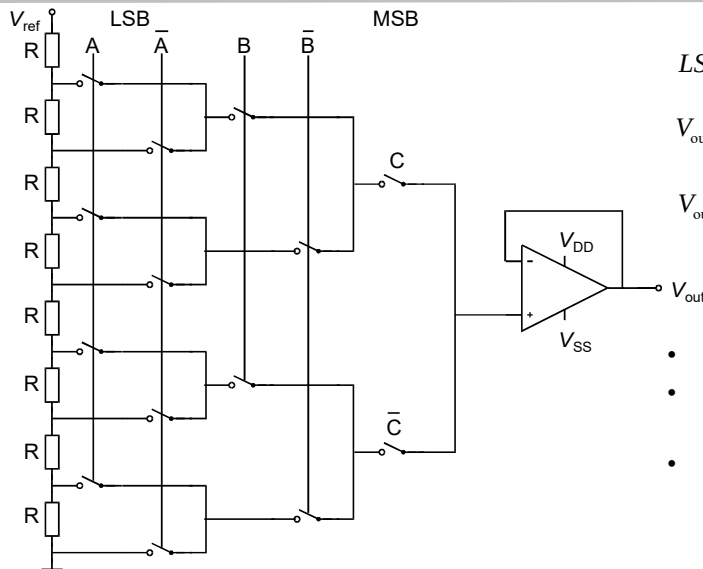
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Example – Resistor String 3-bit DAC



$$LSB = \frac{V_{ref}}{2^B} = \frac{V_{ref}}{8}$$

$$V_{out,max} = \frac{7}{8} V_{ref}$$

$$V_{out,max} = \frac{2^B - 1}{2^B} V_{ref} = V_{ref} - LSB$$

- simple & inherently monotonic
- small area up to ~ 8 bits (256 unit resistors)
- unsuitable for high-resolution, high-speed designs

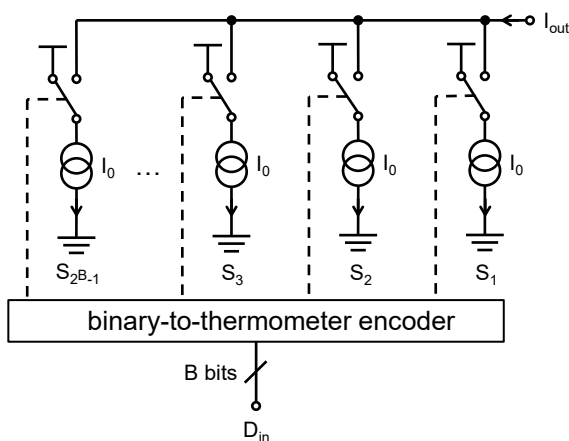
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Current Steering DAC – Thermometer



- inherently monotonic
- large encoder with $2^B - 1$ outputs required
 - impractical for high resolution

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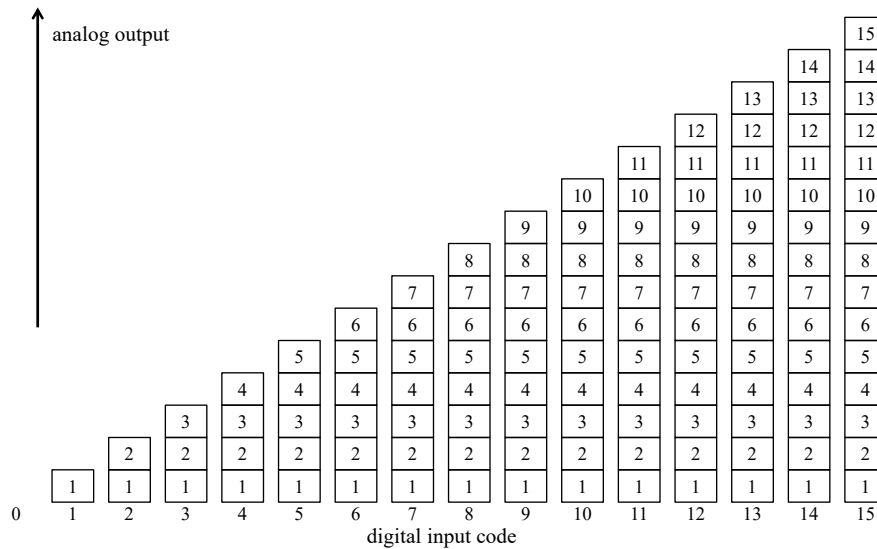
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Thermometer DAC – Principle (4-bit)



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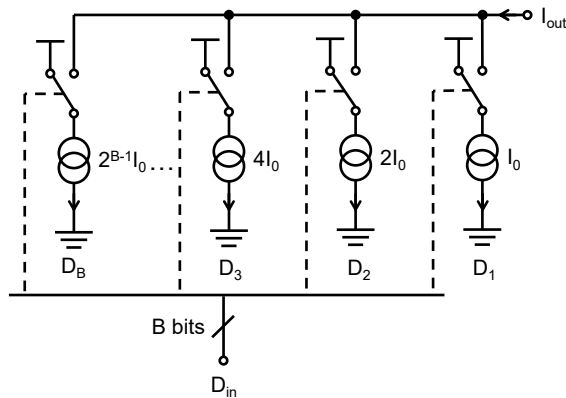
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Current Steering DAC – Binary Weighted



- monotonicity is not guaranteed
- consider transition $100\dots \rightarrow 011\dots$
 - 2^{B-1} source must match sum of others to within 1 LSB to make transition monotonic
- no encoder needed

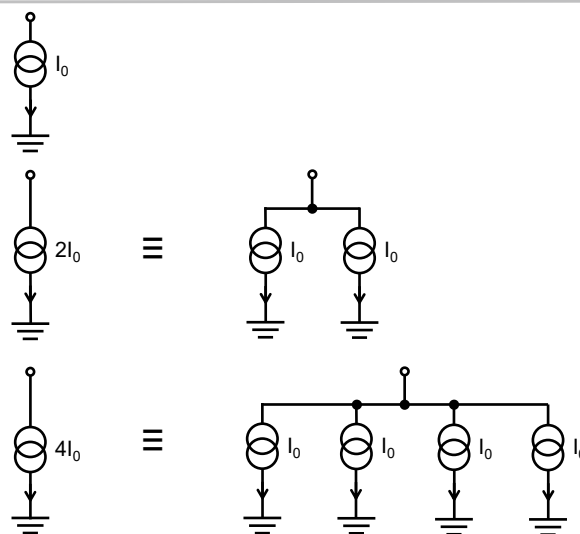
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Implementation of Weighted Elements



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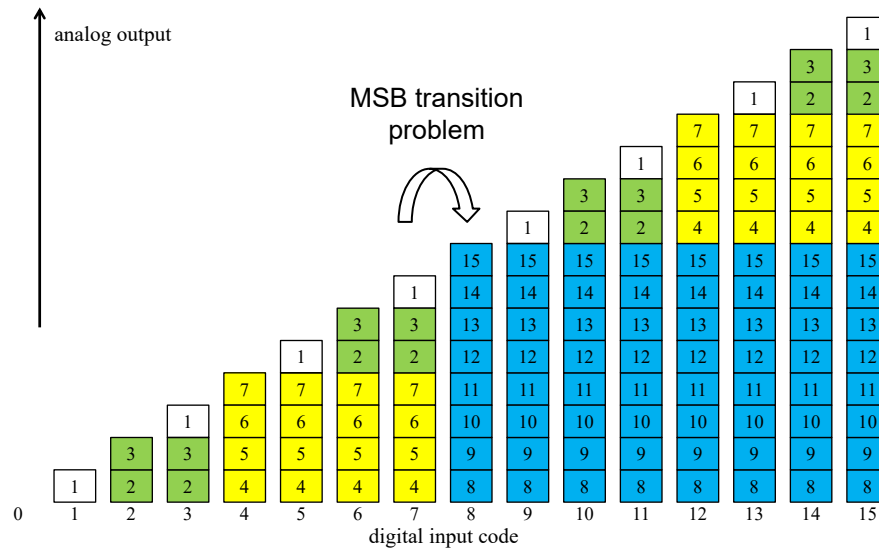
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Binary Weighted DAC – Principle (4-bit)



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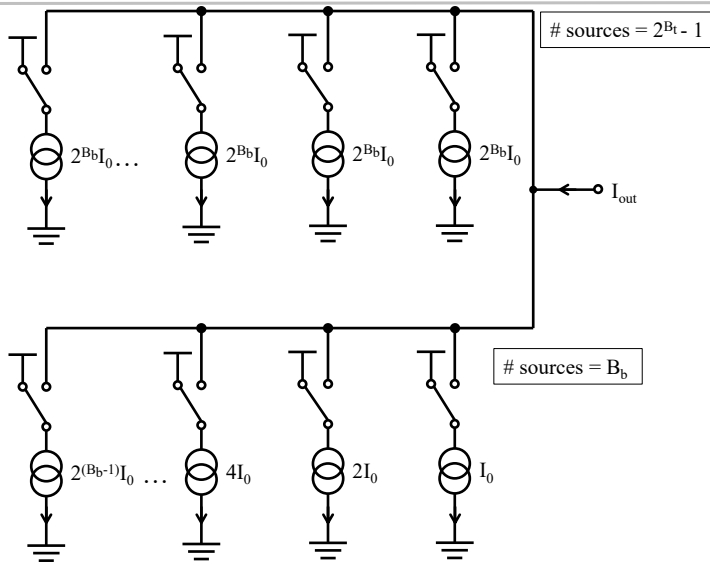
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Current Steering DAC – Segmented



- thermometer MSB section (+ encoder) with $B_t = B - B_b$ bits
- typically $B_t \approx 4 \dots 8$
- easier to achieve monotonicity

- binary weighted LSB section with B_b bits

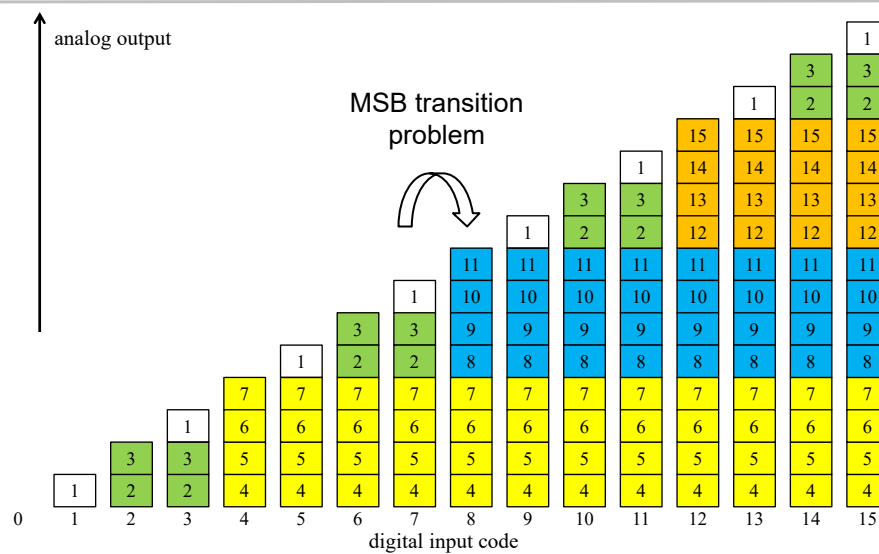
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Segmented DAC – Principle (2-2)



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Overview

- Types
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Static Errors – DNL & INL

- mostly due to unit element mismatch
- systematic errors
 - contact & wiring resistance (IR drop)
 - edge effects in unit element arrays
 - process gradients
 - finite current source output resistance
- random errors
 - lithography
 - often Gaussian distribution (central limit theorem)

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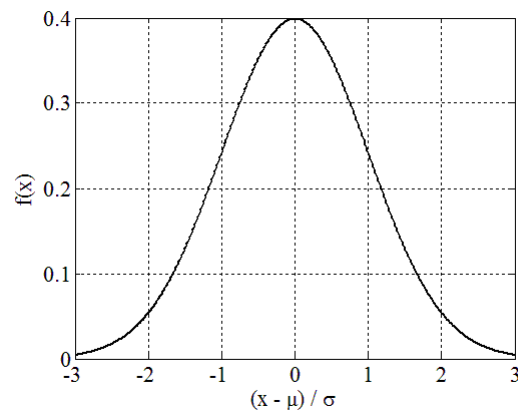
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Gaussian Distribution – Probability Density Function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



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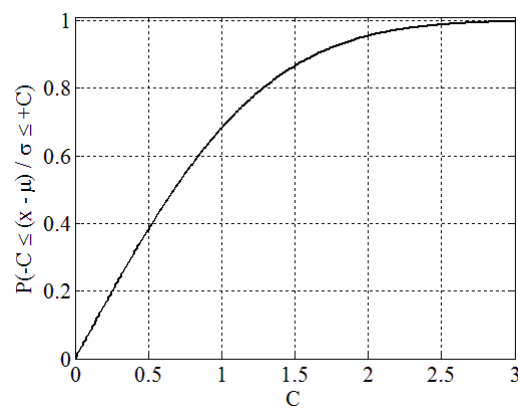
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Yield-Plot

$$P\left(-C \leq \frac{x-\mu}{\sigma} \leq +C\right) = \frac{1}{\sqrt{2\pi}} \int_{-C}^{+C} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \text{erf}\left(\frac{C}{\sqrt{2}}\right)$$



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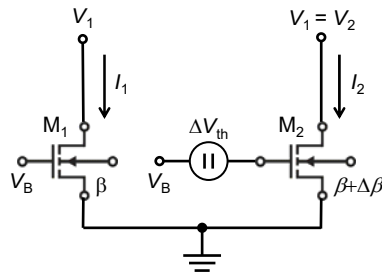
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MOS Current Sources – Mismatch



$$\Delta I = I_1 - I_2 \approx -g_m \Delta V_{th} + I_1 \frac{\Delta \beta}{\beta}$$

$$\frac{\Delta I}{I_1} \approx -\frac{g_m}{I_1} \Delta V_{th} + \frac{\Delta \beta}{\beta}$$

$$\sigma_{\Delta V_{th}} \approx \frac{A_{V_{th}}}{\sqrt{WL}} \quad \sigma_{\frac{\Delta \beta}{\beta}} \approx \frac{A_{\beta}}{\sqrt{WL}}$$

example: $W = 500\mu\text{m}$, $L = 0.2\mu\text{m}$, $g_m/I_D = 10\text{S/A}$, $A_{V_{th}} = 5\text{mV}\mu\text{m}$, $A_{\beta} = 1\%\mu\text{m}$

$$\sigma_{\frac{\Delta I}{I_1}} \approx \sqrt{\left(10 \frac{\text{S}}{\text{A}} \frac{5\text{mV}}{10}\right)^2 + \left(\frac{1\%}{10}\right)^2} = \sqrt{(0.5\%)^2 + (0.1\%)^2} = 0.51\%$$

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Thermometer DAC – Derivation DNL

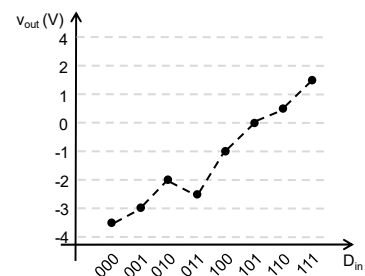
$$DNL(k) = \frac{\text{step}(k) - \text{step}_{\text{avg}}}{\text{step}_{\text{avg}}} = \frac{I_k - I}{I} = \frac{\Delta I}{I}$$

$$\text{stdev}[DNL(k)] = \text{stdev}\left(\frac{\Delta I}{I}\right) = \sigma_u$$

Standard deviation of DNL for each code equals the relative matching σ_u of the unit elements.

Designers tend to rely on simulations rather than on equations

- get rough estimate using simple expressions (often too optimistic)
- run Monte-Carlo simulations to find actual yield or to center specs



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Thermometer DAC – Derivation INL

$$\begin{aligned}
 INL(k) &= \frac{I_{out}(k) - I_{out,uniform}}{step_{avg}} \\
 &= \frac{\sum_{j=1}^k I_j - \frac{k}{N} \sum_{j=1}^N I_j}{\frac{1}{N} \sum_{j=1}^N I_j} = \frac{N \sum_{j=1}^k I_j}{\sum_{j=1}^k I_j + \sum_{j=k+1}^N I_j} - k \rightarrow \\
 &= \frac{A \times N}{A+B} - k
 \end{aligned}$$

$$\begin{aligned}
 var[INL(k)] &= var\left(\frac{A \times N}{A+B} - k\right) \\
 &= N^2 var\left(\frac{A}{A+B}\right) \\
 &= N^2 var\left(\frac{X}{Y}\right)
 \end{aligned}$$

note that $var\left(\frac{X}{Y}\right) \approx \left(\frac{\mu_X}{\mu_Y}\right)^2 \left(\frac{\sigma_X^2}{\mu_X^2} + \frac{\sigma_Y^2}{\mu_Y^2} - 2 \frac{cov(X,Y)}{\mu_X \mu_Y}\right)$

from this approximation it follows

$$var[INL(k)] \approx k \left(1 - \frac{k}{N}\right) \sigma_u^2 \rightarrow \sigma_{INL}(k) \approx \sqrt{k \left(1 - \frac{k}{N}\right)} \sigma_u$$

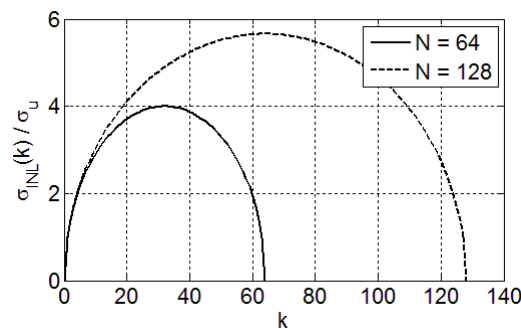
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Thermometer DAC – Plot INL



standard deviation of INL is maximum at mid-scale, i.e., $k = N/2$

$$\sigma_{INL,max} \approx \sqrt{\frac{N}{2} \left(1 - \frac{N/2}{N}\right)} \sigma_u = \frac{1}{2} \sqrt{N} \sigma_u = \frac{1}{2} \sqrt{2^B} \sigma_u$$

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Thermometer DAC – Achievable Resolution

$$\sigma_{INL,max} \approx \frac{1}{2} \sqrt{2^B} \sigma_u \rightarrow B = \log_2 \left(4 \left[\frac{\sigma_{INL,max}}{\sigma_u} \right]^2 \right) = 2 + 2 \log_2 \left(\frac{\sigma_{INL,max}}{\sigma_u} \right)$$

Example: $\sigma_{INL,max} = 0.1\text{LSB}$

σ_u	B
1%	8.6
0.5%	10.6
0.2%	13.3
0.1%	15.3

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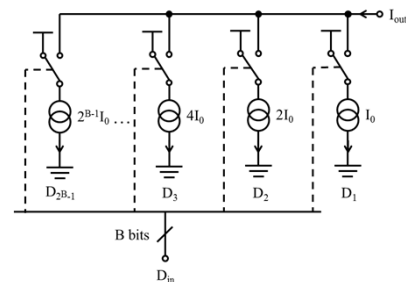
Binary Weighted DAC – DNL & INL

- DNL depends on transition $\rightarrow$ number of switching elements
- worst case: 0111... $\rightarrow$ 1000...
 - turning on MSB & turning off all LSBs

$$\sigma_{DNL}^2 = \underbrace{(2^{B-1} - 1)}_{0111\dots} \sigma_u^2 + \underbrace{(2^{B-1})}_{1000\dots} \sigma_u^2 = (2^B - 1) \sigma_u^2$$

- example
 - $B = 12, \sigma_u = 1\% \rightarrow \sigma_{DNL} = 0.64\text{LSB}$
 - much worse than thermometer DAC ($\sigma_{DNL} = \sigma_u = 0.01\text{LSB}$)

- σ_{INL} same as for thermometer DAC $\rightarrow$ why?



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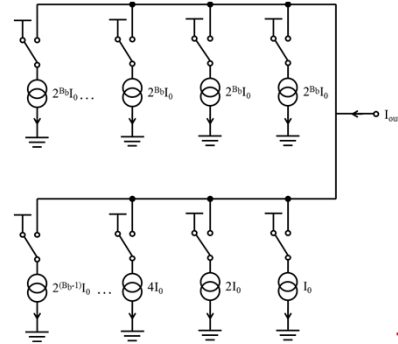
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Segmented DAC – DNL & INL

- DNL depends on transition $\rightarrow$ number of switching elements
- worst case occurs when all sources of the binary section turn off and the first MSB of the thermometer section turns on
 - remember: $B = B_b + B_t$
 $\rightarrow$ equal to an binary weighted DAC with $B_b + 1$ bits
- σ_{INL} same as for thermometer DAC
 $\rightarrow$ why?



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Comparison

	thermometer	segmented	binary weighted
σ_{INL} (worst case)	$\approx \frac{1}{2} \sqrt{2^B} \sigma_u$		
σ_{DNL} (worst case)	$\approx \sigma_u$	$\approx \sqrt{2^{B_b+1} - 1} \sigma_u$	$\approx \sqrt{2^B - 1} \sigma_u$
number of switchable elements	$2^B - 1$	$B_b + 2^{B_t} - 1$	B

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Example – $B = 12$, $\sigma_u = 1\%$

	thermometer	segmented $B_b = 7$, $B_t = 5$	binary weighted
σ_{INL} (worst case)	0.32 LSB		
σ_{DNL} (worst case)	0.01 LSB	0.16 LSB	0.64 LSB
number of switchable elements	4095	38	12

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Overview

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- **Architectures**

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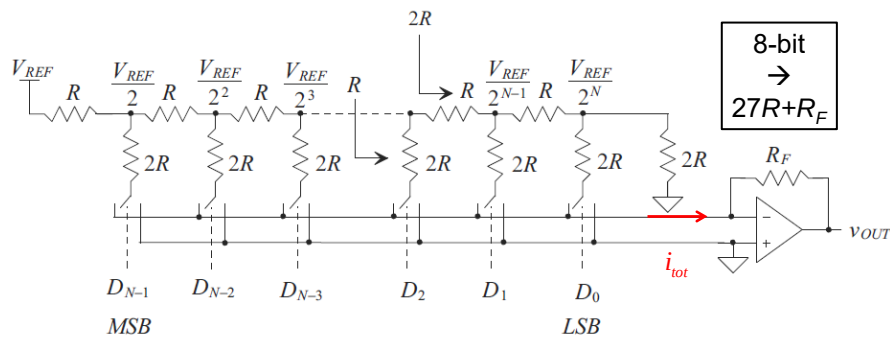
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R-2R Ladder Network



$$v_{out} = -i_{tot} R_F$$

$$i_{tot} = \sum_{k=0}^{N-1} D_k \frac{V_{REF}}{2^{N-k}} \frac{1}{2R}$$

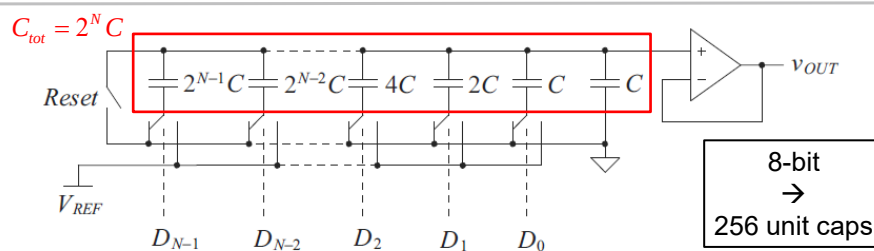
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Charge-Scaling DAC



consider arbitrary yet single $D_k = 1$, i.e., switch k closed, with $0 \leq k \leq N-1$

→ capacitive voltage divider between $C_{tot} - C_k = 2^N C - 2^k C$ and $C_k = 2^k C$

$$\rightarrow v_{out,k} = \frac{\frac{1}{s(2^N - 2^k)C}}{\frac{1}{s2^k C} + \frac{1}{s(2^N - 2^k)C}} V_{REF} = \frac{2^k}{2^N} V_{REF} = 2^{k-N} V_{REF}$$

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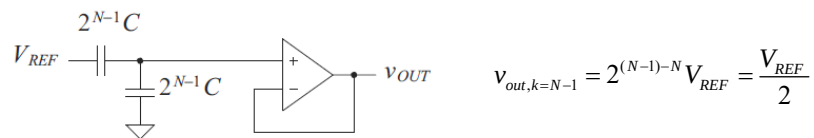
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Charge-Scaling DAC

example: $D_{k=N-1} = 1$, i.e., MSB turned on



→ in order to obtain total output voltage, apply principle of superposition

$$v_{out} = \sum_{k=0}^{N-1} v_{out,k} = \sum_{k=0}^{N-1} D_k 2^{k-N} V_{REF}$$

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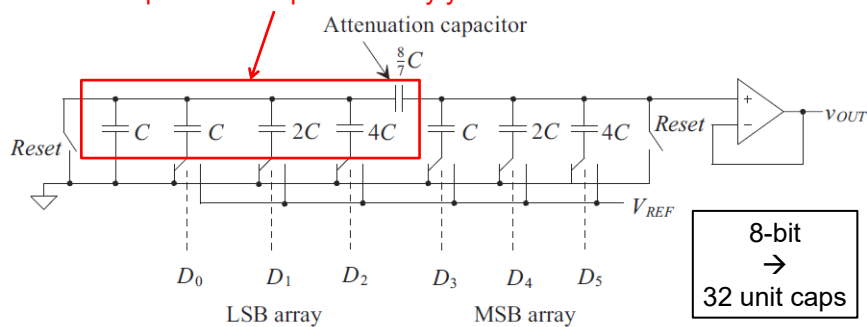
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Charge Scaling DAC – Split Array

series connection of attenuation capacitor
and total capacitor of capacitor array yields C



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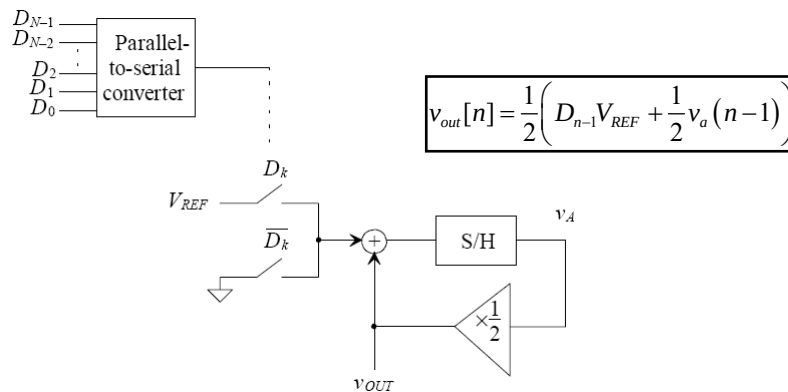
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Cyclic DAC



- N-bits $\rightarrow$ at least N clock cycles required for a complete D/A conversion
 - starts with conversion of LSB, i.e., D_0
 - $\rightarrow$ output after N clock cycles equals $V_{REF}/2^N$

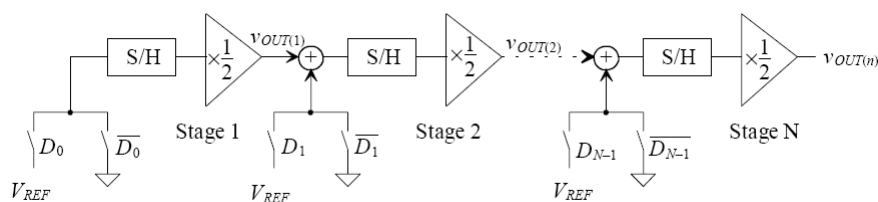
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Pipeline DAC



First D/A conversion requires N clock cycles as signal must make its way through the pipeline.

Second conversion may start after D_0 of first conversion was processed.

$\rightarrow$ After N clock cycles (first conversion ready), conversion is provided every clock cycle.

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